

EXECUTABLE FUNCTIONAL SPECIFICATION

Engineer silicon from a single source of intent.

SyncSilica EFS transforms semiconductor specifications into executable workflows for architecture, design, verification, implementation conformance, and silicon validation.

- 01 **Specification**
- 02 **Architecture intent**
- 03 **Executable flows**
- 04 **Design + verification**
- 05 **EDA implementation**
- 06 **Silicon validation**

The specification layer remains disconnected

Static documents and manual interpretation create duplicated effort, ambiguity, and late discovery.

Static source material

PDF, DOCX, PPT, spreadsheets, tables, and diagrams hold critical intent but are not executable.

Duplicated interpretation

Architecture, RTL, verification, and validation teams independently translate the same requirements.

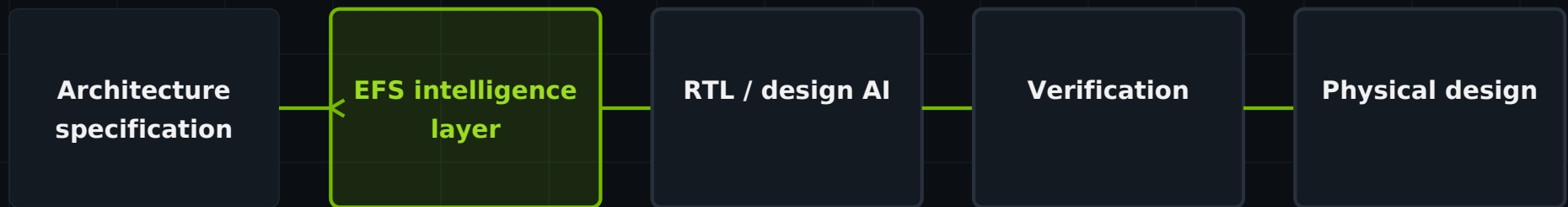
Late mismatch discovery

Ambiguity becomes a verification bug, coverage gap, integration issue, or silicon bring-up failure.

Traditional EDA starts at RTL. EFS begins at specification intent.

EFS introduces the missing architecture intelligence layer

EFS complements RTL copilots and established EDA platforms by supplying executable intent and architecture-derived collateral.



EFS does not replace simulation, synthesis, verification, or physical-design platforms. It creates a traceable executable architecture model that those downstream environments can consume.

What EFS represents

Simple, modular, reusable, analyzable, and testable hardware behavior.

Signals and messages

Values, transitions, opcodes, payloads, and communication across IP instances.

Registers and states

Register events, waits, acknowledgments, state transitions, and state-dependent behavior.

Conditions and timing

Alternate conditions, loops, asynchronous execution, timeouts, and event dependencies.

Hierarchy and reuse

Subsystem flows, reusable subflows, multiple IP instances, and out-of-order sequences.

Specification-driven automation

A control table selects source documents, sections, extraction goals, and optional engineering guidance.

Control table

Document	Section	Description	Comments
compute_SS.docx	Section 5	Generate EFS for I/O	MLOAD/MSTORE
axi_spec.pdf	A3.3	Extract AXI logic	-
axi_spec.pdf	A4.0/A7.0	Additional AXI flows	-

Spec ingestion

Semantic parsing

Protocol / FSM extraction

Executable flow

Generated artifacts

Agent-centric AI

Four coordinated capabilities understand, structure, expand, and execute engineering intent.

Front-end controller

Selects documents, sections, requested output, and optional comments.

Intelligent backend

Extracts events from paragraphs, tables, headers, and references.

Dynamic parser

Maps IF, WHILE, loops, waits, acknowledgments, states, and timing dependencies into logical branches.

Auto expansion

Learns protocol patterns, models overlapping behavior, and generates complete EFS-compatible flows.

Generated engineering assets

Architecture-derived collateral for design, verification, implementation, and validation.

Executable architecture model

Protocol definitions, interface definitions, functional behavior, system requirements, and constraints.

Design collateral

RTL templates, interface modules, register structures, coding guidance, timing constraints, and power intent.

Verification collateral

Testbench scaffolding, assertions, protocol checkers, stimulus, coverage models, and documentation.

Validation collateral

Functional, plug-and-play, volume, cross-feature, and implementation-conformance workflows.

Specification conformance

Prove that implementation behavior matches architecture intent.



AXI proof of concept

Generate, execute, and validate protocol flows from selected specification sections.

Generate

Read/write requests, data, responses, reset, snoop, atomic, and subsystem flows.

Execute

Signal, message, register, subflow, and function events run as an executable model.

Compare

Expected EFS behavior is matched against implementation traces and waveform evidence.

Localize

Exact IF condition, expected value, observed value, and affected transaction are reported.

Demonstrated outcomes: HLFS-to-EFS conversion using agent-centric AI, EFS checker generation, Verilog testbench generation, reset subflow execution, matched AXI transactions, mismatch detection, and volume validation.

Capability comparison

EFS occupies a distinct specification-level position and complements downstream platforms.

Capability	EFS	ChipStack	Cadence	Synopsys	Zero ASIC
Specification parsing	Yes	No	No	No	No
Protocol extraction	Yes	No	No	No	No
Executable specifications	Yes	No	No	No	No
Architecture modeling	Yes	Limited	Limited	Limited	Limited
RTL productivity AI	Limited / roadmap	Yes	Partial	Partial	Partial
Verification automation	Yes	Partial	Yes	Yes	Partial
Spec conformance checking	Yes	No	No	No	No
EDA flow integration	Planned	Yes	Yes	Yes	Yes

ChipStack: AI copilot for RTL productivity. Cadence and Synopsys: broad verification, synthesis, implementation, and signoff platforms. Zero ASIC: cloud-based chip-development platform. EFS: specification parsing, executable intent, and spec-conformance checking.

Industry solutions

Apply executable intent across complex SoC and IP programs.

AI accelerators

Tensor pipelines, DMA, HBM, coherency, command sequencing, error handling, and data movement.

CPU and GPU

Cache-coherent flows, memory subsystems, interconnects, reset, power states, and complex sequencing.

Networking and I/O

AXI, CHI, PCIe, CXL, Ethernet, registers, transactions, and interoperability.

Automotive

ADAS, zonal controllers, BMS, safety behavior, cross-feature interaction, and volume validation.

Datacenter systems

Server fabrics, accelerators, high-speed links, platform initialization, and fleet-scale validation.

Business impact

Improve quality while compressing the semiconductor development cycle.

Business agility

Rapid development of complex SoCs and IP with reusable collateral across product derivatives.

Lower TCO

Automation, earlier validation, and more complete pre-silicon coverage improve execution efficiency.

Traceable intent

Connect requirements, executable flows, generated artifacts, tests, and implementation evidence.

Earlier risk discovery

Expose ambiguity and divergence before late-stage verification or silicon bring-up.

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